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Optimizing Business Performance Through Effective Accounting Information Systems: The Role of System Competence and Information Quality

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Abstract: In today's competitive business environment, accounting information systems (AISs) are crucial for organizations seeking to enhance decision making and improve performance. This study investigates the interplay between AIS competence, information quality, and system effectiveness and their collective impact on business performance within Saudi Arabian companies. Using a quantitative approach, data were collected from 123 manufacturing and service firms through a structured questionnaire. Employing structural equation modeling (SEM), this study elucidates the direct and mediating effects of AIS attributes on organizational outcomes. The findings indicate that system competence has a direct positive effect on both information quality and AIS effectiveness. Information quality, in turn, positively influences AIS effectiveness and business performance. Additionally, AIS effectiveness was found to have a direct positive impact on organizational performance. This study provides valuable insights for managers seeking to optimize AIS investments and emphasizes the importance of integrating high-quality information systems to achieve strategic and operational goals. The results offer a detailed understanding of AIS dynamics, particularly within the context of emerging markets, and contribute to the broader discourse on technology-driven business performance enhancement.

Keywords: accounting information systems (AISs); business performance; system competence; information quality; system effectiveness

1. Introduction

In today's dynamic and competitive business landscape, organizations are increasingly reliant on technology to navigate complex challenges and drive sustainable growth (Hussain et al. 2021). Accounting information systems (AISs), a cornerstone of modern business operations, play a pivotal role in capturing, processing, and disseminating financial and operational data, providing vital insights for informed decision making (O'Brien and Marakas 2006). AIS has evolved from rudimentary systems primarily focused on transaction processing to sophisticated platforms capable of supporting a wide range of functions, including financial reporting, budgeting, inventory management, customer relationship management, and supply chain optimization (Romney and Steinbart 2017). This evolution underscores the growing importance of AIS in enabling organizations to effectively manage resources, enhance efficiency, and gain a competitive edge.

The effectiveness of AIS, however, is not solely dependent on its technological sophistication. Rather, it hinges on a confluence of factors, including the system's competence, the quality of information generated, and the overall effectiveness of its implementation and utilization within the organization (Petter et al. 2013). A well-designed and implemented AIS, characterized by accuracy, timeliness, and relevance, can significantly impact various aspects of business performance, fostering improved operational efficiency, enhanced financial reporting, and more effective strategic decision making (Al-Dmour et al. 2023).

However, poorly designed or inadequately implemented systems can lead to data inaccuracies, inefficiencies, and suboptimal business outcomes (Hertati and Zarkasyi 2015). Therefore, understanding the relationship between the specific attributes of AISs and their impact on business performance is paramount for organizations seeking to leverage technology to achieve their strategic objectives.

Studies have found that firms with more advanced and effective AISs can improve their financial reporting, cost control, resource allocation, and ultimately, their profitability and overall competitiveness (Barney 2000; Dehning and Richardson 2002). Given the importance of AISs in the modern business environment, it is critical to understand the factors that contribute to the successful deployment and use of these systems as well as their impact on organizational outcomes. Investigating this topic will provide valuable insights for both academics and practitioners on how to optimize AISs to enhance business performance.

Despite the growing recognition of the role of accounting information systems in facilitating effective decision making and enhancing business performance, many organizations still struggle with the implementation and optimization of these systems. Prior research has examined the relationship between accounting information systems (AISs) and business performance. However, there is a need for more comprehensive and empirical examinations of the specific factors that drive this relationship. Much of the existing literature has focused on the individual components of AISs, such as information quality or system competence, without considering the synergistic effects of these factors (Ge and Helfert 2013). Furthermore, despite the critical importance of SMEs, most existing studies focus primarily on larger enterprises, leaving a notable gap in understanding how AISs affect performance in smaller firms (Mazzarol 2015). This study aims to address this gap by examining the interplay between AIS competence, information quality, and overall system effectiveness in enhancing business performance within SMEs. The selection of small and medium-sized enterprises (SMEs) for this study is justified by the unique challenges they face, which differ significantly from those of larger corporations. SMEs often operate with constrained resources, limiting their ability to adopt and effectively utilize advanced accounting information systems (Beck et al. 2002). According to Ayyagari et al. (2007), SMEs play a crucial role in economic development, particularly in emerging markets, contributing significantly to employment and innovation. Additionally, research indicates that the impact of AISs on business performance may vary considerably between SMEs and larger firms, largely due to differences in organizational structure and decision-making processes (Johari et al. 2022). By adopting a more holistic approach, this study provides a nuanced understanding of the drivers of AIS effectiveness and their implications for organizational success. The findings offer valuable theoretical and practical insights to guide managers in optimizing their AIS investments and implementations. The primary contributions of this research are threefold. First, it provides empirical evidence on the direct and indirect relationships between AIS attributes and business performance, thus enriching the existing body of literature. Second, it offers practical insights for managers in organizations by highlighting the importance of investing in system competence and information quality to achieve strategic objectives. Finally, this study contributes to the theoretical discourse on AISs by proposing a holistic framework that integrates these attributes, thereby paving the way for future research in the field.

The remainder of this paper is structured as follows. The next section provides a detailed review of the existing literature and develops the research hypotheses on the relationship between AISs and organizational performance, highlighting the key factors and theoretical frameworks that underpin this relationship. The Methodology section outlines the data collection procedures, variable measurements, and analytical techniques employed in this study. The Results section presents the empirical findings. Finally, the Discussion and Conclusions sections interpret the implications of the research findings, discuss the theoretical and practical contributions, and identify areas for future research.

2. Literature Review and Hypotheses Development

Accounting information systems (AISs) have become an integral component of modern business operations, serving as the backbone for financial reporting, decision making, and overall organizational performance (Bouwens and Abernethy 2000; Romney and Steinbart 2017). These computer-based systems collect, store, process, and communicate financial and accounting data, enabling organizations to efficiently manage their resources and make informed strategic decisions (Al-Dhubaibi 2021; Ismail and King 2007; Sajady et al. 2008). The growing reliance on AISs can be attributed to the increasing complexity of the business environment, which is characterized by heightened competition, globalization, and rapidly changing regulations (Soudani 2012; Teru et al. 2017). Effective AISs have been found to enhance various organizational outcomes such as improved financial control, increased productivity, and better resource allocation (Kharuddin et al. 2010; Nicolaou 2000). However, the extent to which an AIS contributes to superior business performance is contingent upon the system's competence, the quality of the information generated, and the overall effectiveness of the AIS implementation (Petter et al. 2013). The effective implementation and utilization of accounting information systems (AISs) is crucial for organizations to achieve their strategic and operational goals. AISs play a vital role in providing timely, accurate, and relevant financial and managerial information to support decision making (Romney and Steinbart 2017). Research has shown that the competence of the AIS, the quality of information it generates, and the overall effectiveness of the system have a significant impact on business performance (Mazzarol 2015).

The role of system competence in enhancing the quality and effectiveness of accounting information systems (AISs) has been a significant area of focus in the recent literature. System competence, defined as the technical and functional capabilities of a system, directly influences the quality of the information produced by an AIS. High system competence ensures accurate, timely, and relevant information, which is crucial for decision-making processes (DeLone and McLean 2003). Research by Algrari and Ahmed (2019) indicated that advanced system features and user-friendly interfaces significantly improve the quality of financial reporting and data reliability. Wang and Liao (2008) investigated the influence of system quality on user satisfaction and information quality. They found that systems with advanced functionalities significantly improve the accuracy and timeliness of information, which is crucial for effective decision making. Additionally, the study by Sedera and Gable (2010) highlighted the role of system competence in facilitating seamless data processing and integration. Their work shows that well-designed systems enhance the reliability and relevance of information, supporting the notion that system competence directly impacts information quality. Moreover, Urbach et al. (2010) analyzed the relationship between system quality, user satisfaction, and information quality. They argued that high system competence, characterized by user-friendly interfaces and robust capabilities, leads to improved information quality by ensuring data accuracy and consistency. Therefore, it is hypothesized that:

H1. *System competence has a direct positive effect on information quality.*

Moreover, system competence contributes to the overall effectiveness of AISs by streamlining accounting processes, reducing errors, and improving efficiency. Studies have shown that systems with high competence levels facilitate better data integration, real-time processing, and comprehensive reporting capabilities, which are essential for effective management and control (Romney and Steinbart 2017). Furthermore, the effectiveness of an AIS is also reflected in its ability to support organizational goals and enhance performance. As noted by Grande et al. (2011), competent systems provide robust support for strategic decision making and operational efficiency. Rom and Rohde (2007) discussed how advanced AIS features enhance decision making by providing accurate and timely information, which is crucial for operational efficiency and strategic planning. Additionally, Boulianne (2014) highlighted the importance of system capabilities in facilitating better

data integration and reporting. This integration is essential for organizations to achieve their goals and improve their overall performance by directly linking system competence with AIS effectiveness. Furthermore, a study by [Elbashir et al. \(2011\)](#) examined the role of business intelligence within AIS, demonstrating that systems with superior technical capabilities support more effective management and control processes. This reflects the growing need for systems to adapt to technological advancements to maintain their effectiveness. Thus, it is hypothesized that:

H2. *System competence has a direct positive effect on the effectiveness of the accounting information system.*

High-quality information, characterized by attributes such as accuracy, completeness, timeliness, and relevance, is crucial for effective decision-making processes within organizations ([DeLone and McLean 1992](#)). Empirical evidence suggests that when businesses have access to high-quality information, they can make more informed strategic decisions, optimize operational processes, and enhance overall efficiency, which in turn positively impacts their performance ([Wixom and Todd 2005](#)). This is particularly relevant in the realm of AISs, where the precision and reliability of financial data play a pivotal role in financial reporting, compliance, and strategic planning. Furthermore, the resource-based view (RBV) theory posits that information quality can be considered a valuable resource that offers a competitive advantage ([Barney 2000](#)). When organizations invest in systems that enhance the quality of their accounting information, they are better positioned to respond to market changes, manage risks, and exploit opportunities more effectively than their competitors ([Petter et al. 2008](#)). Recent studies emphasize the critical role of high-quality data in strategic decision making. For instance, [Bharadwaj et al. \(2013\)](#) highlighted how accurate and timely information from accounting systems enhances operational efficiency and supports competitive strategies. By providing reliable data, firms can make informed decisions that improve productivity and profitability. Additionally, [Ghasemi et al. \(2011\)](#) highlighted the importance of integrating quality information into business processes. They found that organizations leveraging high-quality data experience better alignment of their operations with strategic goals, leading to enhanced overall performance. This is particularly evident in rapidly changing markets where responsive decision making is crucial. Moreover, a study by [Ali and Green \(2012\)](#) demonstrated that enhanced information quality contributes to improved financial outcomes by reducing errors and increasing transparency. This leads to better stakeholder trust and more effective resource management, ultimately boosting business performance.

On the other hand, previous studies highlighted the transformative impact of high-quality data on organizational success. According to [Chen et al. \(2012\)](#), businesses that leverage high-quality information can enhance their decision-making capabilities, leading to improved strategic alignment and operational efficiency. This is crucial in a data-driven business environment where timely and accurate information is a competitive asset. Furthermore, a study by [Popovič et al. \(2012\)](#) emphasized that high information quality in AISs contributes to better business intelligence outcomes, which directly enhance performance metrics such as profitability and market responsiveness. Organizations that invest in improving their information quality are better equipped to adapt to market changes and make proactive decisions, thereby gaining a competitive edge. The work of [McKinney et al. \(2002\)](#) on the impact of information quality on system satisfaction revealed that when users perceive information as being of high quality, their satisfaction with the system increases, encouraging more frequent and effective use. This user satisfaction contributes to more efficient business processes and improved organizational performance, reinforcing the importance of information quality as a strategic resource. This theoretical foundation supports the hypothesis that:

H3. *Information quality has a direct positive effect on business performance.*

A growing body of research has found that the effectiveness of a firm's accounting information system (AIS) can have a significant impact on its overall business performance (Granlund and Mouritsen 2003; Melville et al. 2004). An AIS that is well designed, efficiently managed, and provides high-quality, timely financial and operational data has been shown to support key business processes, enhance decision making, and ultimately drive improvements in metrics like profitability, productivity, and competitive advantage (Ismail and King 2007; Marnewick 2016). For example, a study by Poston and Grabski (2001) found that the use of more effective ERP-based accounting information systems was positively associated with enhanced financial performance measures in a sample of U.S. manufacturing firms. Similarly, Marnewick (2016) revealed that organizations with more effective AISs tended to exhibit higher levels of business agility and responsiveness. AIS implementation can improve decision making and operational efficiency, leading to better business performance (Dandago and Rufai 2014). In addition, organizations with robust AIS frameworks tend to experience higher productivity levels due to enhanced data management and process integration (Salehi et al. 2010). Grande et al. (2011) explored the relationship between AISs and organizational performance in the context of SMEs. Their findings suggest that an effective AIS contributes significantly to improved financial management and competitive advantage, particularly when aligned with business goals. Soudani (2012) demonstrated that integrating advanced digital tools into AISs facilitates better financial reporting and decision making, leading to improved organizational performance. This digital integration allows firms to streamline operations, ultimately boosting productivity and profitability. By improving data accuracy and availability, the AIS supports better strategic planning and resource management. This leads to enhanced operational efficiency and a competitive edge in the market (Al-Mamary et al. 2014). Esmeray (2016) emphasized the importance of data quality and user competence in maximizing AIS benefits. Their findings suggest that firms investing in comprehensive training programs for AIS users experience increased system effectiveness, which translates into improved business outcomes, such as cost reduction and enhanced agility. Building on these findings, the proposed hypothesis posits that:

H4. *Accounting information system effectiveness has a direct positive effect on business performance.*

The quality of information within accounting information systems (AISs) is paramount to their effectiveness. High-quality information, characterized by attributes such as accuracy, timeliness, relevance, and completeness, is essential for decision-making processes and operational efficiency within organizations. Previous research has consistently highlighted that the quality of information significantly impacts the effectiveness of AISs. For instance, Xu (2003) emphasized that accurate and timely information enhances the reliability and usability of accounting systems, thereby improving organizational decision making and performance. Similarly, Nicolaou (2000) found that high-quality information contributes to the overall efficiency and effectiveness of AISs by reducing errors, improving transaction processing, and facilitating better financial reporting. Therefore, it is hypothesized that:

H5a. *Information quality has a direct positive effect on the effectiveness of the accounting information system.*

The effectiveness of an accounting information system (AIS) is critically influenced by the competence of the system, which encompasses its capabilities, functionalities, and adaptability to organizational needs. Prior research indicates that system competence significantly impacts the quality of the information produced (DeLone and McLean 2003). High-quality information, characterized by accuracy, relevance, and timeliness, is essential for effective decision making in accounting processes. Therefore, it can be hypothesized that information quality mediates the relationship between system competence and AIS effectiveness, suggesting that competent systems enhance the quality of information, which in turn improves the overall effectiveness of the AIS (Petter et al. 2008).

H5b. *Information quality mediates the effect of system competence on accounting information system effectiveness.*

Furthermore, AIS effectiveness is crucial for enhancing business performance, as effective systems provide reliable and timely financial information, facilitating better strategic and operational decisions (Romney and Steinbart 2017). The quality of information produced by the AIS is a pivotal factor that determines its effectiveness. High-quality information supports comprehensive reporting and analysis, which are essential for superior business performance (Gelinas et al. 2018). Thus, it can be hypothesized that AIS effectiveness mediates the relationship between information quality and business performance, implying that superior information quality leads to a more effective AIS, which subsequently enhances business performance.

H5c. *Accounting information system effectiveness mediates the effect of information quality on business performance.*

3. Methodology

3.1. Research Design

This research employs a quantitative approach to explore the interplay between accounting system competence (SC), information quality (IQ), accounting information system effectiveness (AISE), and their collective influence on business performance within Saudi Arabian companies. To achieve this objective, a survey research design was adopted, targeting a diverse sample of manufacturing and service firms across the country. A structured questionnaire was carefully developed based on an extensive review of the existing literature. This instrument was also pre-tested to ensure its reliability and validity before data collection. The data collected through the survey were then analyzed using structural equation modeling (SEM) techniques, facilitated by the AMOS 21 software package. This advanced statistical approach allowed the researchers to comprehensively examine the complex relationships between the variables of interest, including potential interaction and mediating effects (Hair et al. 2014; Ho 2006). By adopting this analytical framework, this study was able to move beyond simplistic bivariate analyses and uncover the underlying causal mechanisms through which accounting system competence, information quality, and accounting information system effectiveness contribute to overall business performance and success. The use of SEM enabled the researchers to model the complex, multidimensional nature of the constructs under investigation and assess the direct and indirect pathways through which they influence organizational outcomes (Abdel-Maksoud et al. 2016). This holistic analytical approach provided valuable insights into the intricate dynamics between the key factors and their combined impact on business performance within the Saudi Arabian corporate context.

3.2. Sample and Data

The data for this study were collected from a diverse sample of manufacturing and service companies located in Saudi Arabia. The researchers used a simple random sampling technique to select participants from a comprehensive list of large and medium-sized companies headquartered in Riyadh province, which is recognized as the hub of major Saudi business operations. The questionnaire was directed to the senior accounting, finance, or information systems professionals within each company. These individuals were best positioned to provide accurate and informed responses regarding the accounting information system, its effectiveness, and the overall business performance. In the initial contact with the companies, we requested that the questionnaires be completed by the most appropriate person, such as the chief financial officer, accounting manager, or IT manager, who had direct oversight and understanding of the accounting information systems and processes. Upon receiving the completed questionnaires, we reviewed the respondents' job titles and departments to confirm that they held a relevant position within the orga-

nization and were likely to have the necessary knowledge to provide reliable responses. Of the 274 questionnaires that were distributed, 123 valid responses were obtained, which formed the basis for the subsequent analysis. To mitigate the risk of common method bias, we implemented the procedure as recommended in the literature, e.g., Podsakoff et al. (2003) and Tehseen et al. (2017). The survey instrument assured the respondents of complete anonymity and confidentiality to encourage honest and unbiased responses. The survey questions were designed to be clear, concise, and unambiguous, with the constructs measured using established and validated scales from prior studies.

The questionnaire was structured into three distinct sections. The first section gathered background information about the participating organizations, including their industry and other relevant contextual details. The second section of the questionnaire was designed to evaluate the respondents' perceptions of their organization's overall accounting information system effectiveness. This section also assessed the competence of the accounting software in terms of specific factors such as system security, ease of use, and operational efficiency. Additionally, this section examined the quality of information provided by the accounting information system. The measurement of the system competence construct was adapted from Chang et al. (2012), the information quality construct was adapted from Saha et al. (2012), and the accounting information system effectiveness construct was adapted from Shatat et al. (2013) and (Lam et al. 2014) with the necessary adjustments made for each construct to suit the context of this study. Respondents were asked to indicate their level of agreement with various statements using a seven-point Likert scale, ranging from "strongly disagree" (1) to "strongly agree" (7). The final section of the questionnaire solicited the respondents' assessment of their firm's performance, using both financial and non-financial indicators. The use of self-reported performance measures is a widely accepted approach in the literature, particularly when objective performance data are not readily available to the researcher (Dess and Robinson 1984; Wall et al. 2004). This approach has been adopted in numerous prior studies, e.g., (Henri 2006; Maiga and Jacobs 2008; Tayles et al. 2007; Tsamenyi et al. 2011), supporting its validity and relevance in the current research context.

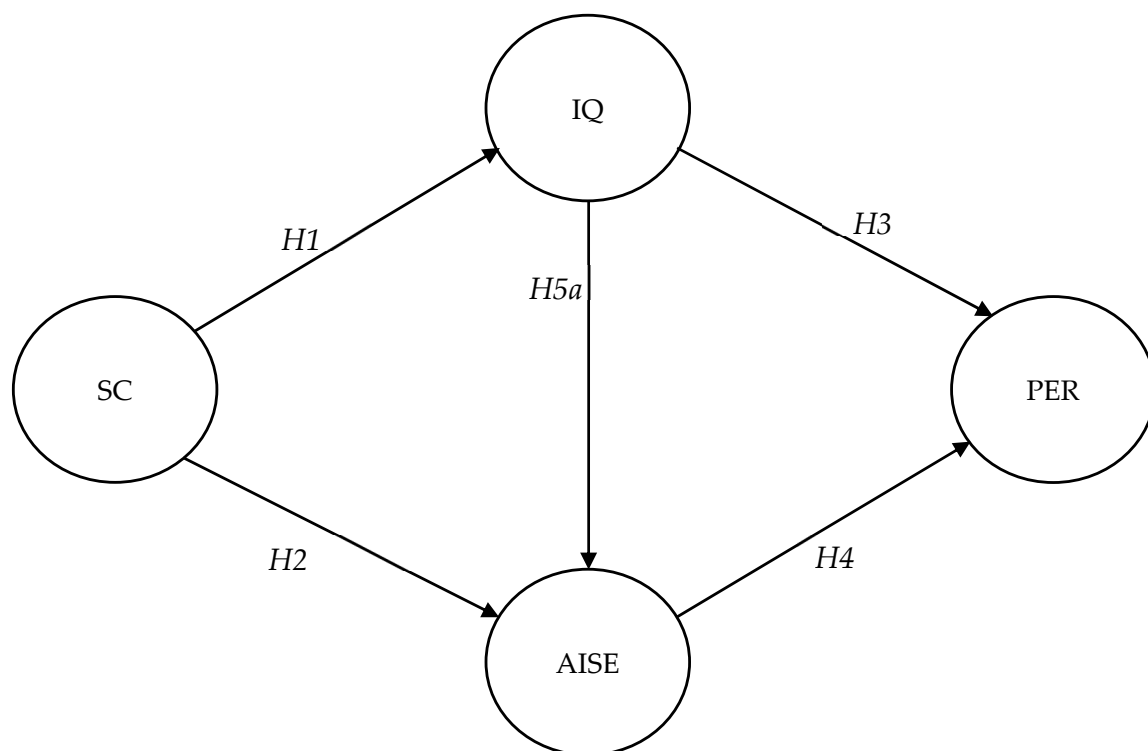
3.3. Path Model

The path model in Figure 1 proposes that system competence and information quality both have a direct effect on the effectiveness of the accounting information system. Additionally, the model suggests a mediating relationship, where SC is hypothesized to influence IQ, which in turn impacts AISE. This implies that system competence may have both a direct effect on AISE and an indirect effect through its influence on information quality. Moreover, information quality and accounting information system effectiveness both have a direct effect on performance. Considering the hypothesized direct effect of IQ on AISE, the model suggests that information quality may have both a direct effect on performance and an indirect effect through its influence on AISE.

To obtain the path coefficients, three structural equations are used:

- (i) $IQ = \beta_0 + \beta_1 SC + \varepsilon$
- (ii) $AISE = \beta_0 + \beta_1 SC + \beta_2 IQ + \varepsilon$
- (iii) $PER = \beta_0 + \beta_1 IQ + \beta_2 AISE + \varepsilon$

where SC = system competence, IQ = information quality, AISE = accounting information system effectiveness, and PER = performance.



H5b: IQ mediates the effect of SC on AISE.

H5c: AISE mediates the effect of IQ on PER.

Figure 1. The theoretical proposed model of the study.

4. Results

4.1. Validation of the Measures

To establish the credibility of the measurement models used in the hypotheses testing, this study scrutinized their reliability and validity. Drawing on the methodological framework proposed by Hooper et al. (2008), the researchers evaluated the unidimensionality and construct validity of each latent variable by assessing the fit of individual constructs. Subsequently, the constructs were integrated into a unified measurement model, which was then subjected to confirmatory factor analysis (CFA) to gauge its overall validity and reliability. To determine the model's capacity to accurately represent the data, the researchers selected a range of goodness-of-fit indices from three primary categories: absolute, incremental, and parsimonious fit measures. In line with Hair et al.'s (2014) recommendation, the researchers chose one index from each category to ensure the model's suitability for further analysis.

Table 1 displays the values obtained for the identified fit indices: one absolute fit index, three incremental fit indices, and one parsimonious fit measure. The actual values for the reported fit indices surpass the established minimum criteria, which are outlined at the top of the table as the accepted benchmark. The accepted benchmarks for each fit index are based on the recommendations by the SEM scholars who recommended values that are >0.90 for GFI (Jöreskog and Sörbom 1984), >0.90 for CFI (Bentler 1990), >0.90 for TLI (Bentler and Bonett 1980), >0.90 for NFI (Bollen 1989), and <5 for ChiSq/df (Marsh and Hocevar 1985).

Table 1. Model fit for measurement models.

Construct	Absolute Fit	Incremental Fit		Parsimonious Fit	
	GFI (>0.90)	CFI (>0.90)	TLI (>0.90)	NFI (>0.90)	ChiSq/df (<5.0)
System Competence	0.903	0.956	0.935	0.936	2.925
Information Quality	0.979	0.996	0.990	0.990	1.594
AIS Effectiveness	0.917	0.969	0.949	0.960	4.076
Performance	0.929	0.965	0.943	0.950	3.264

The discriminant validity of the constructs was evaluated by examining both the correlations between the constructs as well as the square root of the average variance extracted (AVE) for each construct. The findings presented in Table 2 demonstrate that the four constructs exhibit adequate discriminant validity. Specifically, the correlations between each pair of the exogenous latent constructs are below the 0.85 threshold, indicating that they measure distinct concepts. Additionally, the square root of the AVE for each construct is larger than the correlations between that construct and the others. These square roots of the AVE values, shown on the diagonal (in bold) in Table 2, are higher than the corresponding off-diagonal values in the same row and column. This pattern of results provides evidence that the constructs are empirically distinct and capture the unique aspects of the underlying theoretical framework.

Table 2. Discriminant validity index summary.

	System Competence	Information Quality	AIS Effectiveness	Performance
System Competence	0.854			
Information Quality	0.847	0.905		
AIS Effectiveness	0.806	0.805	0.908	
Performance	0.666	0.616	0.609	0.843

The confirmatory factor analysis (CFA) results are summarized in Table 3. The standardized factor loadings and squared multiple correlations (R-squared) for each item across all constructs exceeded the recommended thresholds of 0.60 and 0.40, respectively. This indicates that unidimensionality and construct validity were established. The average variance extracted (AVE) for each construct was calculated by summing the squared standardized factor loadings for a given construct and then dividing that sum by the number of items or indicators for that construct ($AVE = \Sigma(\text{standardized loading}^2)/n$). The AVE measures the amount of variance that a latent construct captures from its indicators relative to the amount of variance due to measurement error. It is a measure of the convergent validity of a construct. An AVE value of 0.50 or higher is generally considered acceptable, indicating that the latent construct explains at least 50% of the variance in its items on average. AVE values below 0.50 suggest that, on average, more variance is due to measurement error than the variance explained by the construct. All the AVE values were found to be greater than 0.5, demonstrating the convergent validity of the constructs. Furthermore, the composite reliability (CR) and Cronbach's alpha values for each construct exceeded the acceptable levels of 0.6 and 0.7, respectively. This suggests that the measures possess adequate internal consistency reliability. Overall, the results of the CFA analysis provide strong evidence for the reliability and validity of the measurement model.

Table 3. Confirmatory factor analysis (CFA) results.

Construct	Item	Factor Loading (>0.6)	R ² (>0.4)	Cronbach's Alpha (>0.7)	CR (>0.6)	AVE (>0.5)
System Competence	SYSC_1	0.89	0.80	0.96	0.96	0.73
	SYSC_2	0.73	0.54			
	SYSC_3	0.87	0.76			
	SYSC_4	0.88	0.77			
	SYSC_5	0.84	0.71			
	SYSC_6	0.86	0.74			
	SYSC_7	0.88	0.77			
	SYSC_8	0.86	0.74			
	SYSC_9	0.86	0.74			
	SYSC_10	0.85	0.73			
Information Quality	INFQ_1	0.88	0.78	0.95	0.95	0.82
	INFQ_2	0.95	0.91			
	INFQ_3	0.93	0.87			
	INFQ_4	0.88	0.78			
	INFQ_5	0.87	0.76			
AIS Effectiveness	AISE_1	0.91	0.82	0.97	0.97	0.83
	AISE_2	0.87	0.76			
	AISE_3	0.90	0.81			
	AISE_4	0.90	0.82			
	AISE_5	0.94	0.89			
	AISE_6	0.92	0.85			
Performance	PER_1	0.77	0.60	0.95	0.95	0.71
	PER_2	0.83	0.68			
	PER_3	0.84	0.71			
	PER_4	0.87	0.76			
	PER_5	0.90	0.81			
	PER_6	0.84	0.71			
	PER_7	0.84	0.71			

4.2. Structural Model Results and Hypotheses Testing

The empirical results of the hypothesis testing are presented in this section. Table 4 shows the direct effects of the exogenous variables on the endogenous variables. The results of the final model analysis indicate that the effect of SC on IQ is statistically significant at the 0.001 level, thereby supporting the first hypothesis (H1) ($\beta = 0.882$, $p < 0.001$). Well-designed and managed systems lead to higher-quality information. Additionally, H2 is supported, with system competence demonstrating a significant positive effect on accounting information system effectiveness (H2: $\beta = 0.423$, $p < 0.001$). The analysis further reveals significant positive direct effects of information quality and accounting information system effectiveness on business performance ($\beta = 0.379$, $p < 0.05$ for H3; $\beta = 0.308$, $p < 0.05$ for H4). These findings concur with the established literature that demonstrates the positive impact of high-quality information on business outcomes and highlights the role of effective accounting information systems in enhancing performance.

Table 4. The standardized and regression coefficients (β) of the structural model.

Hypothesized Paths			Standardized Estimate	Regression Estimate	S.E.	C.R.	p-Value
System Competence	--->	Information Quality	0.882	0.805	0.067	12.080	***
System Competence	--->	AIS Effectiveness	0.423	0.391	0.123	3.181	0.001
Information Quality	--->	AIS Effectiveness	0.432	0.438	0.135	3.239	0.001
Information Quality	--->	Performance	0.379	0.304	0.109	2.796	0.005
AIS Effectiveness	--->	Performance	0.308	0.243	0.106	2.302	0.021

*** Significant at the $p < 0.001$ level.

Finally, the analysis confirms the expected positive effect of information quality on the accounting information system effectiveness, supporting H5a. However, the hypothesized mediating effects (H5b and H5c) are not supported. While information quality has a significant direct effect on accounting information system effectiveness, it does not mediate the relationship between system competence and AISE. Table 5 shows that the direct effect

of SC on AISE is significant ($\beta = 0.423, p < 0.001$) and stronger than the indirect effect (SC on IQ $\times$ IQ on AISE, $\beta = 0.381$). Similarly, the effectiveness of the accounting information system does not mediate the effect of information quality on business performance. The results presented in Table 5 show that the direct effect of IQ on performance is significant ($\beta = 0.379, p < 0.05$) and is stronger than the indirect effect through AISE ($\beta = 0.133$). The model fit statistics indicate a good fit with the data, explaining a substantial portion of the variance in information quality ($R^2 = 0.779$), accounting information system effectiveness ($R^2 = 0.687$), and business performance ($R^2 = 0.425$).

Table 5. The results of the direct and the indirect effect analysis (standardized coefficients).

Hypothesized Paths				Direct Effect	Indirect Effect	Test of the Mediation
SC	----->		AISE	0.423 ***		No Mediation
SC	--->	IQ	AISE		$(0.882 \text{ ***}) \times (0.432 \text{ ***})$ $= 0.381$	
IQ	----->		PER	0.379 **		No Mediation
IQ	--->	AISE	PER		$(0.432 \text{ ***}) \times (0.308 \text{ ***})$ $= 0.133$	

*** Significant at the $p < 0.001$ level. ** Significant at the $p < 0.05$ level.

The results of the hypothesized model are visually represented in Figure 2, which provides a comprehensive overview of the relationships between the variables. Specifically, the figure illustrates the direct and indirect relationships between the independent and dependent variables as well as the magnitude of these relationships through the standardized path coefficients. Additionally, the figure reports the squared multiple correlation coefficients (R^2), which quantify the amount of variance in the dependent variables that can be attributed to the collective influence of the independent variables. In essence, R^2 provides a measure of the predictive power of the model, indicating the proportion of variance in the outcome variables that is explained by the predictor variables.

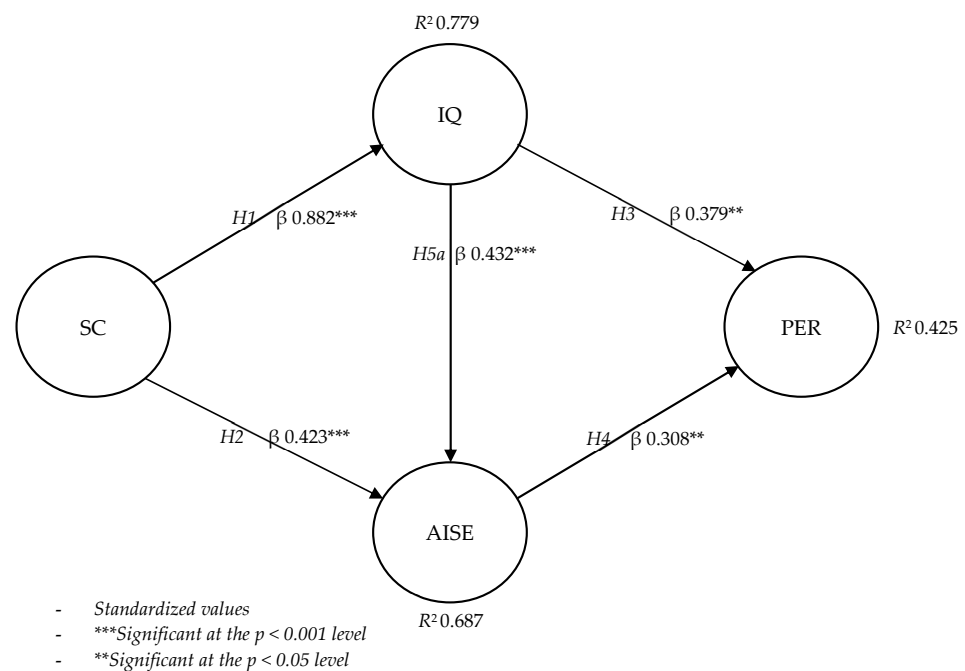


Figure 2. Path diagram with path coefficients from the results of the structural equation modeling.

5. Discussion

The findings of this study provide robust support for the proposition that system competence has a direct positive effect on both information quality and accounting information system (AIS) effectiveness. These results align with those of previous studies, such as those by DeLone and McLean (1992) and Seddon (1997), who identified system competence as a critical determinant of system success. Specifically, this study reaffirms the notion that higher levels of system competence, encompassing the technical skills and knowledge required to manage and use information systems, significantly enhance the quality of the information produced. This is consistent with the work of Wixom and Todd (2005), who emphasized that technical capabilities are essential for generating accurate, timely, and relevant information, which are the key dimensions of information quality. Moreover, our findings extend the understanding of system competence by demonstrating its substantial impact on the overall effectiveness of AISs. This insight is particularly important because it addresses a gap in the literature where the direct relationship between system competence and AIS effectiveness has been less explored. Prior research, such as that by Nicolaou (2000), primarily focused on the indirect effects of system competence through user satisfaction and system usage. By establishing a direct positive link, our study contributes to a more nuanced understanding of how technical expertise directly influences the operational success of AISs, which includes improved decision making, enhanced control, and better financial reporting. In comparison with previous studies, our research provides empirical evidence that strengthens the theoretical framework surrounding system competence and its outcomes. For instance, our findings support the assertions made by Petter et al. (2013) on the importance of technical competence in information system success models. However, unlike past studies, which often treated system competence as a background variable or an indirect influencer, our study highlights its direct impact, thereby offering a more comprehensive perspective. This contribution is particularly valuable for practitioners and policymakers aiming to enhance AIS performance, as it underscores the importance of investing in technical training and development to achieve superior information quality and system effectiveness.

The results of this study provide compelling evidence supporting the proposition that both information quality and accounting information system (AIS) effectiveness have a direct positive impact on business performance. These findings align with the existing literature, such as DeLone and McLean's (1992) information systems success model, which underscores the importance of high-quality information in enhancing organizational performance. Our study corroborates the assertion by McLeod and Schell (2006) that accurate, timely, and relevant information significantly enhances decision-making processes, leading to better business outcomes. Moreover, the positive relationship between AIS effectiveness and business performance is consistent with findings from past research. For instance, studies by Nicolaou (2000) and Grande et al. (2011) demonstrated that effective AIS implementations streamline financial reporting, enhance operational efficiency, and support strategic management processes, which collectively boost overall business performance.

The consistency of our results with these previous studies reinforces the critical role of AISs in facilitating superior business outcomes through improved financial management and operational processes. However, this study contributes a novel perspective by integrating these two dimensions, information quality and AIS effectiveness, into a unified framework for assessing their combined impact on business performance. While prior studies have often treated these factors independently, the current research highlights the synergistic effects of high-quality information and an effective AIS on enhancing business outcomes. This integrated approach provides a more comprehensive understanding of the mechanisms through which accounting information systems contribute to organizational success, offering valuable insights for both academics and practitioners aiming to optimize AIS implementations for improved business performance. The findings of this study underscore the pivotal role of information quality in enhancing the effectiveness of accounting information systems (AISs). These findings align with prior research by DeLone

and McLean (2003) and Gorla et al. (2010). These studies have consistently highlighted that high-quality information, characterized by accuracy, completeness, relevance, and timeliness, is crucial for the operational success and decision-making capabilities of an AIS. Our results emphasize the importance of prioritizing information quality to maximize the utility and effectiveness of AISs, thereby reinforcing the established understanding within the field.

Hypothesis H5b suggests that information quality mediates the effect of system competence on AIS effectiveness. Contrary to our expectations, the analysis did not support this mediation effect, indicating that while system competence is essential, its impact on AIS effectiveness may not be significantly channeled through information quality alone. This finding differs from that of Nicolaou (2000), who suggested a strong interplay between system competence and information quality in determining AIS outcomes. Our study contributes to the literature by suggesting that there might be other mediating factors or direct paths through which system competence influences AIS effectiveness, warranting further exploration.

Similarly, hypothesis H5c, which proposed that AIS effectiveness would mediate the relationship between information quality and business performance, was not supported. This result contrasts with the assertions made by Seddon (1997), who argued for a direct correlation between high-quality information, effective AISs, and improved business performance. The absence of this mediating effect in our study highlights the complexity of the relationship between AIS effectiveness and business performance, suggesting that other variables, such as organizational culture or external conditions, might play a more substantial role. Thus, our study provides novel insights by challenging the linear assumptions in previous models and encouraging a more nuanced understanding of the determinants of business performance in the context of AISs.

In general, the findings of this study align with previous research that emphasizes the significance of system competence and information quality in enhancing AIS effectiveness and, ultimately, organizational performance. For instance, DeLone and McLean (2003) and Seddon (1997) highlighted system competence as a critical determinant of system success. However, the current study extends this understanding by demonstrating the direct and significant impact of system competence on information quality and AIS effectiveness. Prior studies, such as that by Nicolaou (2000), mainly focused on the indirect effects of system competence through user satisfaction and system usage. By establishing a direct link, the current study contributes a more nuanced understanding of how technical expertise directly influences the operational success of AISs.

Furthermore, while previous research has primarily focused on the individual components of AISs, such as information quality or system competence, the present study examines the interplay between these factors and their combined impact on business performance. This integrative approach offers a more comprehensive understanding of the mechanisms through which accounting information systems contribute to organizational success, enriching the existing literature and providing valuable insights for both academics and practitioners. Notably, the study aligns with the assertions made by Petter et al. (2013) on the importance of technical competence in information system success models, but it extends this understanding by directly examining the impact of system competence on information quality and AIS effectiveness. This emphasis on the direct impact of system competence is a significant contribution that offers a more comprehensive perspective and provides valuable guidance for practitioners and policymakers aiming to enhance AIS performance.

6. Conclusions

This study examined a comprehensive model of the hypothesized relationships within the domain of accounting information systems and their impact on business performance. Employing structural equation modeling analysis using the AMOS software package, the investigation yielded substantive empirical support for most proposed theoretical

linkages. Notably, the findings indicated that system competence, characterized by the technical capabilities and overall performance of the accounting information system, exerts a direct, positive influence on both the quality of information produced and the broader effectiveness of the system itself. This suggests that the core technological underpinnings of the accounting information infrastructure serve as an essential foundation for realizing the benefits of high-caliber financial data and an efficient, responsive system.

Moreover, the results illustrated the critical role of information quality in driving positive business outcomes. Specifically, the study found that the accuracy, relevance, and timeliness of the accounting information have a direct, positive effect on overall business performance. This underscores the strategic importance of investing in systems and processes that ensure the integrity and usefulness of the financial data relied upon by organizational decision makers. Complementing this, the investigation also revealed that the effectiveness of the accounting information system itself—encompassing factors such as system flexibility, integration, and user satisfaction—exhibits a direct positive relationship with business performance. This finding highlights the practical significance of optimizing the design, implementation, and ongoing management of accounting information systems to maximize their contribution to firm-level success. Further elucidating these dynamics, this study supports the hypothesis that information quality exerts a direct, positive influence on the overall effectiveness of the accounting information system. This implies a reciprocal, reinforcing relationship between the quality of the data produced and the system's capacity to efficiently and reliably deliver that information to organizational stakeholders. However, the study did not find support for the hypothesized mediation effects. Specifically, information quality does not mediate the relationship between system competence and AIS effectiveness, nor does AIS effectiveness mediate the relationship between information quality and business performance.

In summary, the findings emphasize the critical roles of system competence, information quality, and AIS effectiveness in directly enhancing business performance. However, the anticipated mediating roles of information quality and AIS effectiveness were not validated in this study, suggesting that their impacts on business performance are more straightforward and less interdependent than initially hypothesized. This insight provides valuable guidance for organizations aiming to optimize their accounting information systems to boost overall business performance.

7. Implications, Limitations, and Future Research

The findings of this study offer several practical implications for organizations seeking to enhance their business performance through improved accounting information systems. First, the direct positive effects of system competence on both information quality and AIS effectiveness suggest that investing in skilled personnel and robust system capabilities is crucial. Organizations should prioritize training and development programs to enhance the competence of their system users and ensure that the systems in place are well designed and efficient. Additionally, since information quality and AIS effectiveness directly impact business performance, companies should implement rigorous data management practices and regularly evaluate their AIS to ensure it meets evolving business needs. By focusing on these areas, organizations can achieve more reliable and actionable insights, leading to better decision making and improved business outcomes.

Despite its contributions, this study has several limitations that should be acknowledged. First, the research is based on a survey, which could imply the subjectivity of the respondents in evaluating their companies' AISs. Future research could employ a case study methodology. Case studies allow for an in-depth examination of the relationships between system competence, information quality, AIS effectiveness, and business performance within specific organizational contexts. By focusing on individual organizations or specific sectors, researchers can explore how these variables interact over time and in different circumstances, thereby offering a more comprehensive understanding of the causal mechanisms at play. Future research might also investigate the influence of other

relevant variables such as user acceptance, data security, and organizational culture on system competence, information quality, AIS effectiveness, and business performance.

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References

- Abdel-Maksoud, Ahmed, Walid Cheffi, and Kilani Ghoudi. 2016. The mediating effect of shop-floor involvement on relations between advanced management accounting practices and operational non-financial performance indicators. *The British Accounting Review* 48: 169–84. [\[CrossRef\]](#)
- Al-Dhubaibi, Ahmed Abdullah Saad. 2021. Modeling Managerial Accounting Information Systems Acceptance and Intention of Retention: Activity based Costing System as an Example. *WSEAS Transactions on Business and Economics* 18: 1461–73. [\[CrossRef\]](#)
- Al-Dmour, Ahmed, Hala Zaidan, and Abdul Rahman Al Natour. 2023. The impact knowledge management processes on business performance via the role of accounting information quality as a mediating factor. *VINE Journal of Information Knowledge Management Systems* 53: 523–43. [\[CrossRef\]](#)
- Algrari, Ahmed Yass, and Mr. Rebwar Mohammed Ahmed. 2019. The impact of Accounting Information Systems' Quality on Accounting Information Quality. *Journal of Information Technology Management* 11: 62–80.
- Ali, Syaiful, and Peter Green. 2012. Effective information technology (IT) governance mechanisms: An IT outsourcing perspective. *Information Systems Frontiers* 14: 179–93. [\[CrossRef\]](#)
- Al-Mamary, Yaser Hasan, Alina Shamsuddin, and Nor Aziati. 2014. The role of different types of information systems in business organizations: A review. *International Journal of Research* 1: 333–39.
- Ayyagari, Meghana, Thorsten Beck, and Asli Demircug-Kunt. 2007. Small and medium enterprises across the globe. *Small Business Economics* 29: 415–34. [\[CrossRef\]](#)
- Barney, Jay B. 2000. Firm resources and sustained competitive advantage. In *Economics Meets Sociology in Strategic Management (Advances in Strategic Management)*. Leeds: Emerald Group Publishing Limited, pp. 203–27.
- Beck, Thorsten, Asli Demircug Kunt, and Vojislav Maksimovic. 2002. *Financial and Legal Constraints to Firm Growth Does Size Matter?* Washington, DC: Banco Mundial.
- Bentler, Peter M. 1990. Comparative fit indexes in structural models. *Psychological Bulletin* 107: 238. [\[CrossRef\]](#)
- Bentler, Peter M., and Douglas G. Bonett. 1980. Significance tests and goodness of fit in the analysis of covariance structures. *Psychological Bulletin* 88: 588. [\[CrossRef\]](#)
- Bharadwaj, Anandhi, Omar A. El Sawy, Paul A. Pavlou, and N. Venkatraman. 2013. Digital business strategy: Toward a next generation of insights. *MIS Quarterly* 2013: 471–82. [\[CrossRef\]](#)
- Bollen, Kenneth A. 1989. A new incremental fit index for general structural equation models. *Sociological Methods & Research* 17: 303–16.
- Boulianne, Emilio. 2014. Impact of accounting software utilization on students' knowledge acquisition: An important change in accounting education. *Journal of Accounting & Organizational Change* 10: 22–48.

- Bouwens, Jan, and Margaret A. Abernethy. 2000. The consequences of customization on management accounting system design. *Accounting, Organizations and Society* 25: 221–41. [\[CrossRef\]](#)
- Chang, Ching-Sheng, Su-Yueh Chen, and Yi-Ting Lan. 2012. Motivating medical information system performance by system quality, service quality, and job satisfaction for evidence-based practice. *BMC Medical Informatics and Decision Making* 12: 1–12. [\[CrossRef\]](#)
- Chen, Hsinchun, Roger H. L. Chiang, and Veda C. Storey. 2012. Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, 1165–88.
- Dandago, Kabiru I., and Abdullahi Sani Rufai. 2014. Information technology and accounting information system in the Nigerian banking industry. *Asian Economic and Financial Review* 4: 655–70.
- Dehning, Bruce, and Vernon J. Richardson. 2002. Returns on investments in information technology: A research synthesis. *Journal of Information Systems* 16: 7–30. [\[CrossRef\]](#)
- DeLone, William H., and Ephraim R. McLean. 1992. Information systems success: The quest for the dependent variable. *Information Systems Research* 3: 60–95. [\[CrossRef\]](#)
- DeLone, William H., and Ephraim R. McLean. 2003. The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems* 19: 9–30.
- Dess, Gregory G., and Richard B. Robinson. 1984. Measuring organizational performance in the absence of objective measures: The case of the privately-held firm and conglomerate business unit. *Strategic Management Journal* 5: 265–73. [\[CrossRef\]](#)
- Elbashir, Mohamed Z., Philip A. Collier, and Steve G. Sutton. 2011. The role of organizational absorptive capacity in strategic use of business intelligence to support integrated management control systems. *The Accounting Review* 86: 155–84. [\[CrossRef\]](#)
- Esmeray, Azize. 2016. The impact of accounting information systems (AIS) on firm performance: Empirical evidence in Turkish small and medium sized enterprises. *International Review of Management and Marketing* 6: 233–36.
- Ge, Mouzhi, and Markus Helfert. 2013. Impact of information quality on supply chain decisions. *Journal of Computer Information Systems* 53: 59–67. [\[CrossRef\]](#)
- Gelinas, Ulric J., Richard B. Dull, and Patrick Wheeler. 2018. *Accounting Information Systems*. Southbank: Cengage AU.
- Ghasemi, Maziyar, Vahid Shafeiepour, Mohammad Aslani, and Elham Barvayeh. 2011. The impact of Information Technology (IT) on modern accounting systems. *Procedia-Social and Behavioral Sciences* 28: 112–16. [\[CrossRef\]](#)
- Gorla, Narasimhaiah, Toni M. Somers, and Betty Wong. 2010. Organizational impact of system quality, information quality, and service quality. *The Journal of Strategic Information Systems* 19: 207–28. [\[CrossRef\]](#)
- Grande, Elena Urquía, Raquel Pérez Estébanez, and Clara Munoz Colomina. 2011. The impact of Accounting Information Systems (AIS) on performance measures: Empirical evidence in Spanish SMEs. *The International Journal of Digital Accounting Research* 11: 25–43.
- Granlund, Markus, and Jan Mouritsen. 2003. Special section on management control and new information technologies. *European Accounting Review* 12: 77–83. [\[CrossRef\]](#)
- Hair, Joseph F., William C. Black, Barry J. Babin, and Rolph E. Anderson. 2014. *Multivariate Data Analysis: Pearson New International Edition*. Harlow: Pearson Education Limited.
- Henri, Jean-François. 2006. Organizational culture and performance measurement systems. *Accounting, Organizations and Society* 31: 77–103. [\[CrossRef\]](#)
- Hertati, Lesi, and P. Zarkasyi. 2015. Effect of Competence User Information System, The Quality of Accounting Information Systems Management and Implications Insatisfaction User Information System (State Owner in Sumatera Selatan). *European Journal of Accounting, Auditing and Finance Research* 3: 35–60.
- Ho, Robert. 2006. *Handbook of Univariate and Multivariate Data Analysis and Interpretation with SPSS*. Boca Raton: Taylor & Francis Group, Chapman & Hall/CRC.
- Hooper, Daire, Joseph Coughlan, and Michael Mullen. 2008. Structural equation modelling: Guidelines for determining model fit. *Electronic Journal of Business Research Methods* 6: 53–60.
- Hussain, Hafezali Iqbal, Fakarudin Kamarudin, Zuraidah Mohd-Sanusi, Shuhaida Mohamed Shuhidan, Ahmed Abdullah Saad Al-Dhubaibi, and Mohd Shahril Ahmad Razimi. 2021. Governance in the internet of vehicles (IoV) context: Examination of information privacy, transport anxiety, intention, and usage. *Journal of Advanced Transportation* 2021: 5563260. [\[CrossRef\]](#)
- Ismail, Noor Azizi, and Malcolm King. 2007. Factors influencing the alignment of accounting information systems in small and medium sized Malaysian manufacturing firms. *Journal of Information Systems and Small Business* 1: 1–20.
- Johari, Nor Hasimah, Nurul Nazwa Khairudin, Nadhirah Mohd Rasidi, Iezza Syaida Yuhana, and Nurusyafiq Atiqah Norbadir. 2022. Accounting Information System and Organizational Effectiveness: Evidence From Smes Manufacturing Companies. Paper presented at the International Symposium & Exhibition on Business and Accounting 2022 (ISEBA 2022), Pahang, Malaysia, 28 September 2022.
- Jöreskog, Karl G., and Dag Sörbom. 1984. *Analysis of Linear Structural Relationships by Maximum Likelihood, Instrumental Variables, and Least Squares Methods*. Mooresville: Scientific Software.
- Kharuddin, Saira, Zariyawati Mohd Ashhari, and Annuar Md Nassir. 2010. Information system and firms' performance: The case of Malaysian small medium enterprises. *International Business Research* 3: 28–35.
- Lam, Thanh Hien, Thanh-Lam Than, and Cuong Pham. 2014. Key determinants of information system effectiveness: An empirical case in Lac Hong University. *International Journal of Information Technology and Business Management* 32: 1–14.

- Maiga, Adam S., and Fred A. Jacobs. 2008. Extent of ABC Use and Its Consequence. *Contemporary Accounting Research* 25: 533–66. [\[CrossRef\]](#)
- Marnewick, Carl. 2016. Benefits of information system projects: The tale of two countries. *International Journal of Project Management* 34: 748–60. [\[CrossRef\]](#)
- Marsh, Herbert W., and Dennis Hocevar. 1985. Application of confirmatory factor analysis to the study of self-concept: First-and higher order factor models and their invariance across groups. *Psychological Bulletin* 97: 562. [\[CrossRef\]](#)
- Mazzarol, Tim. 2015. SMEs engagement with e-commerce, e-business and e-marketing. *Small Enterprise Research* 22: 79–90. [\[CrossRef\]](#)
- McKinney, Vicki, Kanghyun Yoon, and Fatemeh Mariam Zahedi. 2002. The measurement of web-customer satisfaction: An expectation and disconfirmation approach. *Information Systems Research* 13: 296–315. [\[CrossRef\]](#)
- McLeod, Raymond, and George Schell. 2006. *Management Information Systems*. London: Pearson.
- Melville, Nigel, Kenneth Kraemer, and Vijay Gurbaxani. 2004. Information technology and organizational performance: An integrative model of IT business value. *Management Information Systems Quarterly* 28: 7. [\[CrossRef\]](#)
- Nicolaou, Andreas I. 2000. A contingency model of perceived effectiveness in accounting information systems: Organizational coordination and control effects. *International Journal of Accounting Information Systems* 1: 91–105. [\[CrossRef\]](#)
- O'Brien, James A., and George M. Marakas. 2006. *Management Information Systems*. New York: McGraw-Hill Irwin.
- Petter, Stacie, William DeLone, and Ephraim McLean. 2008. Measuring information systems success: Models, dimensions, measures, and interrelationships. *European Journal of Information Systems* 17: 236–63. [\[CrossRef\]](#)
- Petter, Stacie, William DeLone, and Ephraim R. McLean. 2013. Information systems success: The quest for the independent variables. *Journal of Management Information Systems* 29: 7–62. [\[CrossRef\]](#)
- Podsakoff, Philip M., Scott B. MacKenzie, Jeong-Yeon Lee, and Nathan P. Podsakoff. 2003. Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology* 88: 879. [\[CrossRef\]](#)
- Popovič, Aleš, Ray Hackney, Pedro Simões Coelho, and Jurij Jaklič. 2012. Towards business intelligence systems success: Effects of maturity and culture on analytical decision making. *Decision Support Systems* 54: 729–39. [\[CrossRef\]](#)
- Poston, Robin, and Severin Grabski. 2001. Financial impacts of enterprise resource planning implementations. *International Journal of Accounting Information Systems* 2: 271–94. [\[CrossRef\]](#)
- Rom, Anders, and Carsten Rohde. 2007. Management accounting and integrated information systems: A literature review. *International Journal of Accounting Information Systems* 8: 40–68. [\[CrossRef\]](#)
- Romney, Marshall B., and Paul John Steinbart. 2017. *Accounting Information System*. London: Pearson Education Limited.
- Saha, Parmita, Atanu K. Nath, and Esmail Salehi-Sangari. 2012. Evaluation of government e-tax websites: An information quality and system quality approach. *Transforming Government: People, Process and Policy* 6: 300–21. [\[CrossRef\]](#)
- Sajady, Hussein, Mohsen Dastgir, and Hashem Hashem Nejad. 2008. Evaluation of the effectiveness of accounting information systems. *International Journal of Information Science and Management* 6: 49–59.
- Salehi, Mahdi, Vahab Rostami, and Abdolkarim Mogadam. 2010. Usefulness of accounting information system in emerging economy: Empirical evidence of Iran. *International Journal of Economics and Finance* 2: 186–95. [\[CrossRef\]](#)
- Seddon, Peter B. 1997. A respecification and extension of the DeLone and McLean model of IS success. *Information Systems Research* 8: 240–53. [\[CrossRef\]](#)
- Sedera, Darshana, and Guy G. Gable. 2010. Knowledge management competence for enterprise system success. *The Journal of Strategic Information Systems* 19: 296–306. [\[CrossRef\]](#)
- Shatat, Abdallah S., Zawiyah Mohd Yusof, and J. Abd Aziz. 2013. The impact of information system success on business intelligence system effectiveness. *Journal of Theoretical and Applied Information Technology* 50: 512–22.
- Soudani, Siamak Nejadhosseini. 2012. The usefulness of an accounting information system for effective organizational performance. *International Journal of Economics and Finance* 4: 136–45. [\[CrossRef\]](#)
- Tayles, Mike, Richard Pike, and Saudah Sofian. 2007. Intellectual capital, management accounting practices and corporate performance: Perceptions of managers. *Accounting, Auditing & Accountability Journal* 20: 522–48.
- Tehseen, Shehnaz, T. Ramayah, and Sulaiman Sajilan. 2017. Testing and controlling for common method variance: A review of available methods. *Journal of Management Sciences* 4: 142–68. [\[CrossRef\]](#)
- Teru, Susan Peter, Innocent Idoku, and Jane Tinyang Ndeyati. 2017. A review of the impact of accounting information system for effective internal control on firm performance. *Indian Journal of Finance and Banking* 1: 52–59. [\[CrossRef\]](#)
- Tsamenyi, Mathew, Sunil Sahadev, and Zheng Shi Qiao. 2011. The relationship between business strategy, management control systems and performance: Evidence from China. *Advances in Accounting* 27: 193–203. [\[CrossRef\]](#)
- Urbach, Nils, Stefan Smolnik, and Gerold Riempp. 2010. An empirical investigation of employee portal success. *The Journal of Strategic Information Systems* 19: 184–206. [\[CrossRef\]](#)
- Wall, Toby D., Jonathan Michie, Malcolm Patterson, Stephen J. Wood, Maura Sheehan, Chris W. Clegg, and Michael West. 2004. On the validity of subjective measures of company performance. *Personnel Psychology* 57: 95–118. [\[CrossRef\]](#)
- Wang, Yi-Shun, and Yi-Wen Liao. 2008. Assessing eGovernment systems success: A validation of the DeLone and McLean model of information systems success. *Government Information Quarterly* 25: 717–33. [\[CrossRef\]](#)

- Wixom, Barbara H., and Peter A. Todd. 2005. A theoretical integration of user satisfaction and technology acceptance. *Information Systems Research* 16: 85–102. [[CrossRef](#)]
- Xu, Hongjiang. 2003. Critical Success Factors for Accounting Information Systems Data Quality. Ph.D. thesis, University of Southern Queensland, Springfield, QLD, Australia.

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